MSG No. 194.268 $P6'_3/m'm'c$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-3m'`.

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,7,8,9,16,17,18,22,23,24]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[4,5,6,10,11,12,13,14,15,19,20,21]`

Table 2: Wyckoff site: `2b`, site symmetry: `-6'm'2`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,4,5,6,19,20,21,22,23,24]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[7,8,9,10,11,12,13,14,15,16,17,18]`

Table 3: Wyckoff site: `2c`, site symmetry: `-6'm'2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[1,2,3,4,5,6,19,20,21,22,23,24]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[7,8,9,10,11,12,13,14,15,16,17,18]`

Table 4: Wyckoff site: `2d`, site symmetry: `-6'm'2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[1,2,3,4,5,6,19,20,21,22,23,24]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[7,8,9,10,11,12,13,14,15,16,17,18]`

Table 5: Wyckoff site: `4e`, site symmetry: `3m'`.

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,22,23,24]`
2	$\left[0, 0, \frac{1}{2} - z\right]$	`[4,5,6,19,20,21]`
3	$[0, 0, -z]$	`[7,8,9,16,17,18]`
4	$\left[0, 0, z + \frac{1}{2}\right]$	`[10,11,12,13,14,15]`

Table 6: Wyckoff site: `4f`, site symmetry: `3m'`.

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,2,3,22,23,24]`
2	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	`[4,5,6,19,20,21]`
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[7,8,9,16,17,18]`
4	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	`[10,11,12,13,14,15]`

Table 7: Wyckoff site: `6g`, site symmetry: `.2'/m'.`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,7,16,22]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,8,17,23]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,9,18,24]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,10,13,19]`
5	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,11,14,20]`
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[6,12,15,21]`

Table 8: Wyckoff site: `6h`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[x, 2x, \frac{1}{4}\right]$	`[1,5,20,22]`
2	$\left[-2x, -x, \frac{1}{4}\right]$	`[2,6,21,23]`
3	$\left[x, -x, \frac{1}{4}\right]$	`[3,4,19,24]`
4	$\left[-x, -2x, \frac{3}{4}\right]$	`[7,11,14,16]`
5	$\left[2x, x, \frac{3}{4}\right]$	`[8,12,15,17]`
6	$\left[-x, x, \frac{3}{4}\right]$	`[9,10,13,18]`

Table 9: Wyckoff site: `12i`, site symmetry: `.2'.`

No.	position	mapping
1	$[x, 0, 0]$	`[1,16]`
2	$[0, x, 0]$	`[2,17]`
3	$[-x, -x, 0]$	`[3,18]`
4	$\left[x, x, \frac{1}{2}\right]$	`[4,13]`
5	$\left[-x, 0, \frac{1}{2}\right]$	`[5,14]`
6	$\left[0, -x, \frac{1}{2}\right]$	`[6,15]`
7	$[-x, 0, 0]$	`[7,22]`
8	$[0, -x, 0]$	`[8,23]`
9	$[x, x, 0]$	`[9,24]`
10	$\left[-x, -x, \frac{1}{2}\right]$	`[10,19]`
11	$\left[x, 0, \frac{1}{2}\right]$	`[11,20]`
12	$\left[0, x, \frac{1}{2}\right]$	`[12,21]`

Table 10: Wyckoff site: `12j`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x, y, \frac{1}{4}\right]$	`[1,20]`
2	$\left[-y, x-y, \frac{1}{4}\right]$	`[2,21]`
3	$\left[-x+y, -x, \frac{1}{4}\right]$	`[3,19]`
4	$\left[x, x-y, \frac{1}{4}\right]$	`[4,24]`
5	$\left[-x+y, y, \frac{1}{4}\right]$	`[5,22]`
6	$\left[-y, -x, \frac{1}{4}\right]$	`[6,23]`
7	$\left[-x, -y, \frac{3}{4}\right]$	`[7,14]`

continued ...

Table 10

No.	position	mapping
8	$\left[y,\, -x+y,\, \frac{3}{4}\right]$	[8,15]
9	$\left[x-y,\, x,\, \frac{3}{4}\right]$	[9,13]
10	$\left[-x,\, -x+y,\, \frac{3}{4}\right]$	[10,18]
11	$\left[x-y,\, -y,\, \frac{3}{4}\right]$	[11,16]
12	$\left[y,\, x,\, \frac{3}{4}\right]$	[12,17]

Table 11: Wyckoff site: 12k, site symmetry: .m'.

No.	position	mapping
1	$[x,\, 2x,\, z]$	[1,22]
2	$[-2x,\, -x,\, z]$	[2,23]
3	$[x,\, -x,\, z]$	[3,24]
4	$\left[x,\, -x,\, \frac{1}{2}-z\right]$	[4,19]
5	$\left[x,\, 2x,\, \frac{1}{2}-z\right]$	[5,20]
6	$\left[-2x,\, -x,\, \frac{1}{2}-z\right]$	[6,21]
7	$[-x,\, -2x,\, -z]$	[7,16]
8	$[2x,\, x,\, -z]$	[8,17]
9	$[-x,\, x,\, -z]$	[9,18]
10	$\left[-x,\, x,\, z+\frac{1}{2}\right]$	[10,13]
11	$\left[-x,\, -2x,\, z+\frac{1}{2}\right]$	[11,14]
12	$\left[2x,\, x,\, z+\frac{1}{2}\right]$	[12,15]

Table 12: Wyckoff site: 24l, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-y,\, x-y,\, z]$	[2]
3	$[-x+y,\, -x,\, z]$	[3]
4	$\left[x,\, x-y,\, \frac{1}{2}-z\right]$	[4]
5	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	[5]
6	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[6]
7	$[-x,\, -y,\, -z]$	[7]
8	$[y,\, -x+y,\, -z]$	[8]
9	$[x-y,\, x,\, -z]$	[9]
10	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	[10]
11	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	[11]
12	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[12]
13	$\left[x-y,\, x,\, z+\frac{1}{2}\right]$	[13]
14	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[y,\, -x+y,\, z+\frac{1}{2}\right]$	[15]
16	$[x-y,\, -y,\, -z]$	[16]
17	$[y,\, x,\, -z]$	[17]
18	$[-x,\, -x+y,\, -z]$	[18]
19	$\left[-x+y,\, -x,\, \frac{1}{2}-z\right]$	[19]

continued ...

Table 12

No.	position	mapping
20	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[20]
21	$\left[-y,\, x-y,\, \frac{1}{2}-z\right]$	[21]
22	$[-x+y,\, y,\, z]$	[22]
23	$[-y,\, -x,\, z]$	[23]
24	$[x,\, x-y,\, z]$	[24]